



MINISTRY OF HIGHER EDUCATION
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Smart Online Class Management Mobile Application

For Wisdom Educational Unit

2nd Year 2nd Semester Project Proposal

(Final Project)

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1. Introduction

The education sector has rapidly evolved with the advancement of digital technology and online learning systems. Many educational institutes still rely on manual processes and social media platforms such as WhatsApp and Facebook to manage online classes, communicate with students, and share educational materials. These methods often create difficulties in managing educational activities efficiently.

Wisdom Educational Unit currently conducts educational classes for students using traditional communication and management methods. Students experience difficulties accessing class schedules, assignments, study materials, and announcements through a centralized system. Teachers and administrators also face challenges in managing academic activities effectively.

Therefore, this project proposes the development of a Smart Online Class Management Mobile Application that provides a centralized platform for students, teachers, and administrators. The system will improve online class management, communication, assignment handling, attendance tracking, and educational resource sharing.

The application will be developed using Flutter and Firebase technologies to provide a secure, scalable, and user-friendly mobile solution.

2. Aim & Objectives

Aim

To develop a smart mobile application that improves online class management and communication processes at Wisdom Educational Unit – Aluthgama.

Objectives

- i) **Student Management:** Provide secure student registration, login, and profile management functionalities.
- ii) **Online Class Management:** Allow teachers and administrators to create and manage online classes with schedules and meeting links.
- iii) **Assignment & Note Sharing:** Enable teachers to upload assignments, notes, and educational resources for students.
- iv) **Notification System:** Provide real-time notifications for class schedules, announcements, and assignment deadlines.
- v) **Teacher Management:** Allow teachers to manage classes, monitor student activities, and upload learning materials.
- vi) **Attendance & Progress Tracking:** Provide attendance and student progress tracking functionalities.
- vii) **Admin Dashboard:** Enable administrators to manage students, teachers, classes, and institute activities efficiently.
- viii) **Mobile Accessibility:** Provide easy access to educational resources through Android mobile devices anytime and anywhere.

3. Functional Requirements

The system should provide the following functional requirements.

I. User Authentication

- The system shall allow students, teachers, and administrators to register accounts.
- The system shall provide secure login and logout functionality.
- The system shall allow users to reset forgotten passwords.
- The system shall authenticate users according to their roles.

II. Student Management

- The system shall allow students to create and update profiles.
- The system shall allow students to enroll in classes.
- The system shall allow students to view class schedules.
- The system shall allow students to access notes and learning materials.
- The system shall allow students to submit assignments.
- The system shall allow students to receive notifications and announcements.

III. Teacher Management

- The system shall allow teachers to create and manage classes.
- The system shall allow teachers to upload study materials and assignments.
- The system shall allow teachers to publish announcements.
- The system shall allow teachers to monitor student attendance and performance.

IV. Admin Management

- The system shall allow administrators to manage students and teachers.
- The system shall allow administrators to manage classes and schedules.
- The system shall allow administrators to generate reports.
- The system shall monitor overall system activities.

V. Assignment Management

- The system shall allow teachers to upload assignments with deadlines.
- The system shall allow students to upload assignment submissions.
- The system shall maintain assignment records securely.

VI. Notification System

- The system shall send notifications regarding classes and announcements.
- The system shall notify users about assignment deadlines.
- The system shall provide real-time updates.

VII. Attendance Management

- The system shall allow teachers to mark student attendance.
- The system shall store attendance records in the database.
- The system shall allow administrators to monitor attendance reports.

4. Non-Functional Requirements

I. Performance Requirements

- The system should load pages within a few seconds.
- The system should support multiple users simultaneously.
- The application should provide real-time database updates.

II. Security Requirements

- User passwords should be encrypted securely.
- Only authorized users should access specific modules.
- The system should prevent unauthorized access.

III. Usability Requirements

- The application should provide a simple and user-friendly interface.
- Users should navigate the application easily.
- Error messages should be clear and understandable.

IV. Reliability Requirements

- The system should operate continuously with minimal downtime.
- The application should provide accurate and consistent information.

V. Scalability Requirements

- The system should support future growth of users and classes.
- The application should support future feature enhancements.

VI. Compatibility Requirements

- The application should support Android devices.
- The system should work properly on different screen sizes.

VII. Backup & Recovery Requirements

- The system should maintain secure data backups.
- The application should support data recovery during failures.

5. Business Process

I. User Registration & Authentication

Students, teachers, and administrators register and log into the system securely using Firebase Authentication. User information is stored in Firebase Firestore Database.

II. Class Management

Teachers create classes by adding class details, schedules, subject information, and online meeting links. Students can enroll and access classes through the mobile application.

III. Study Material Upload

Teachers upload notes, assignments, and study materials in PDF, image, or document formats. Students can access and download educational materials through the application.

IV. Assignment Submission

Students submit assignments directly through the mobile application. Teachers review submissions and provide feedback.

V. Notification & Announcements

The system sends notifications regarding upcoming classes, assignment deadlines, and institute announcements.

VI. Attendance & Progress Monitoring

Teachers and administrators monitor student attendance and academic progress through the system dashboard.

6. Scope of Project

The system will cover:

- Student registration and authentication
- Teacher and admin management
- Online class scheduling
- Study material sharing
- Assignment management
- Notifications and announcements
- Student attendance tracking
- Firebase real-time database integration
- Mobile application development using Flutter

Limitations:

- Internet connection is required to access the system.
- Payment gateway integration is not included in the initial version.
- Video conferencing will use external platforms such as Zoom Platform.
- The system mainly focuses on educational management activities.

7. Problem Definition & Proposed Solutions

Problems:

- Educational activities are managed manually.
- No centralized system exists for online learning management.
- Communication between teachers and students is inefficient.
- Assignments and notes are shared through multiple platforms.
- Attendance and progress tracking are difficult.

Proposed Solution:

- Develop a centralized mobile application for educational management.
- Provide secure login and role-based access.
- Enable efficient sharing of assignments, notes, and announcements.
- Implement attendance and notification systems.
- Improve communication and academic management efficiency.

8. Technologies Used

Task	Purpose
Flutter	Mobile application development
Dart	Programming language used with Flutter
Firebase Authentication	User login and authentication
Firebase Firestore	Real-time cloud database
Firebase Storage	File and document storage
Firebase Cloud Messaging	Notification management
Android Studio	Development environment
GitHub	Version control and source code management
Figma	UI/UX design and prototyping

Technology Description:

Flutter

Flutter is used to develop modern and responsive Android mobile applications.

Dart

Dart is the programming language used to implement system functionalities.

Firebase

Firebase provides authentication, database management, cloud storage, and notification services.

Android Studio

Android Studio is used for coding, testing, and application deployment.

GitHub

GitHub is used for project version control and source code backup.

Figma

Figma is used to design application interfaces and wireframes.

9. Timeline

Task	Weeks														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Requirement Gathering															
UI/UX Design															
Database Design															
Authentication System															
Student Module Development															
Teacher Module Development															
Admin Module Development															
Testing & Bug Fixing															
Final Documentation															
Deployment & Presentation															

10. Conclusion

The Smart Online Class Management Mobile Application will provide an efficient and modern solution for managing educational activities at Wisdom Educational Unit – Aluthgama. The system will improve communication, simplify academic management processes, and provide centralized access to educational resources for students, teachers, and administrators.

Furthermore, the project demonstrates practical implementation of mobile application development, database management, authentication systems, and software engineering concepts using modern technologies such as Flutter and Firebase.

Overall, the system will contribute toward improving digital education management and enhancing the learning experience within the institute.

11. References

- <https://flutter.dev/docs>
- <https://firebase.google.com/docs>
- <https://developer.android.com>
- <https://dart.dev/guides>
- <https://git-scm.com/doc>
- <https://github.com>
- <https://www.figma.com/community>

12. Declaration

I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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Signature of the Supervisor

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Date